DO NOW — RECALL

Testing and program quality

Name: _____ Class: _____ Date: _____

1. Why should Largest() require a nonempty input array?

2. What is a regression?

3. What does a class method do?

4. A date validator passes all unit tests but booking fails when writing to the database. What test is needed?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

It reads a[0] as its starting value.

2. Model answer

Previously working functionality fails after a change.

3. Model answer

Defines behaviour for objects of that class.

4. Model answer

An integration test of the database interaction.

Vocabulary — know these terms

unit testing — Test one module/function in isolation.

integration testing — Check interaction of components.

system testing — Test the entire application as a system.

acceptance testing — Check fitness for user/business requirements.

regression testing — Repeat tests to catch defects after changes.

NEW LEARNING

Read, annotate and explain

Test individual units

Unit tests isolate one function or class method from the rest of the system.

Test connections and behaviour

Integration tests interactions between modules; system tests the complete application.

Confirm intended outcome

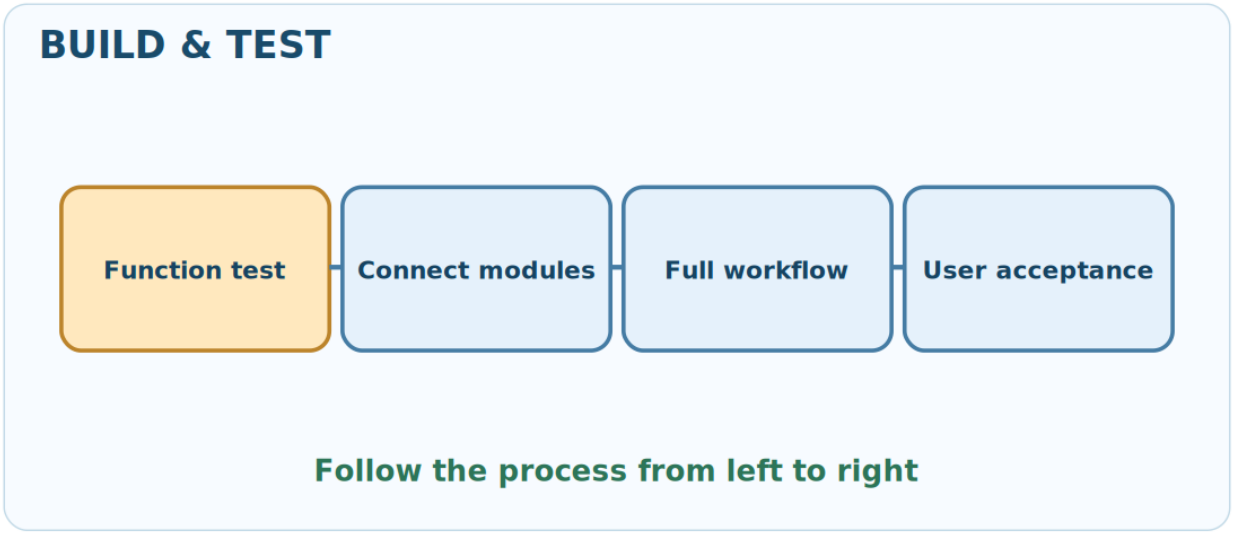
Acceptance testing checks whether the delivered application meets stakeholder requirements. Automated regression tests rerun after changes.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

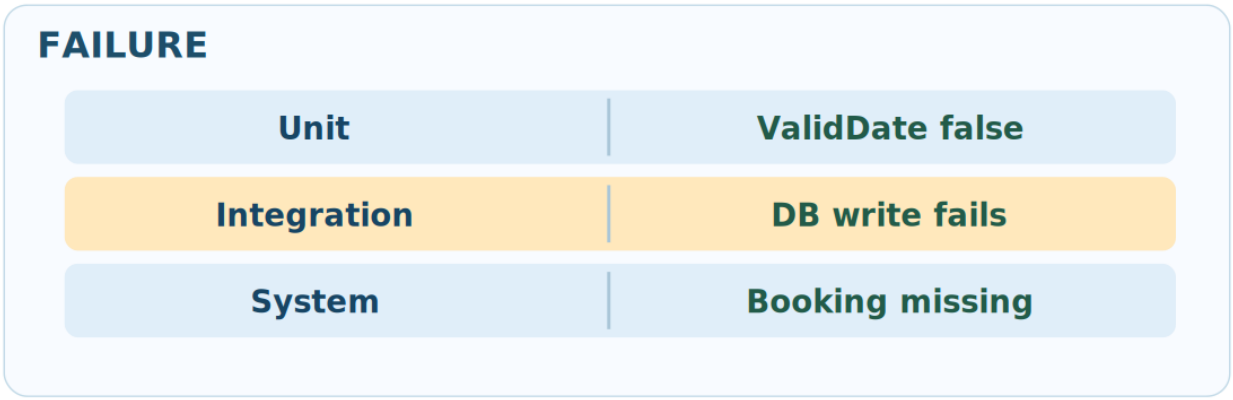
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



BUILD & TEST; Function test to Connect modules to Full workflow to User acceptance

New Learning



FAILURE: Unit ValidDate false; Integration DB write fails; System Booking missing

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



REGRESSION LOOP; Make change to Run tests to Investigate to Fix

I DO — WORKED EXAMPLE

Follow the worked method

A checkout computes the right total but fails to save orders. Which test is most relevant?

Predict first

TotalCalculator already gives correct values; the failure appears when OrderService interacts with the database.

Model and explain

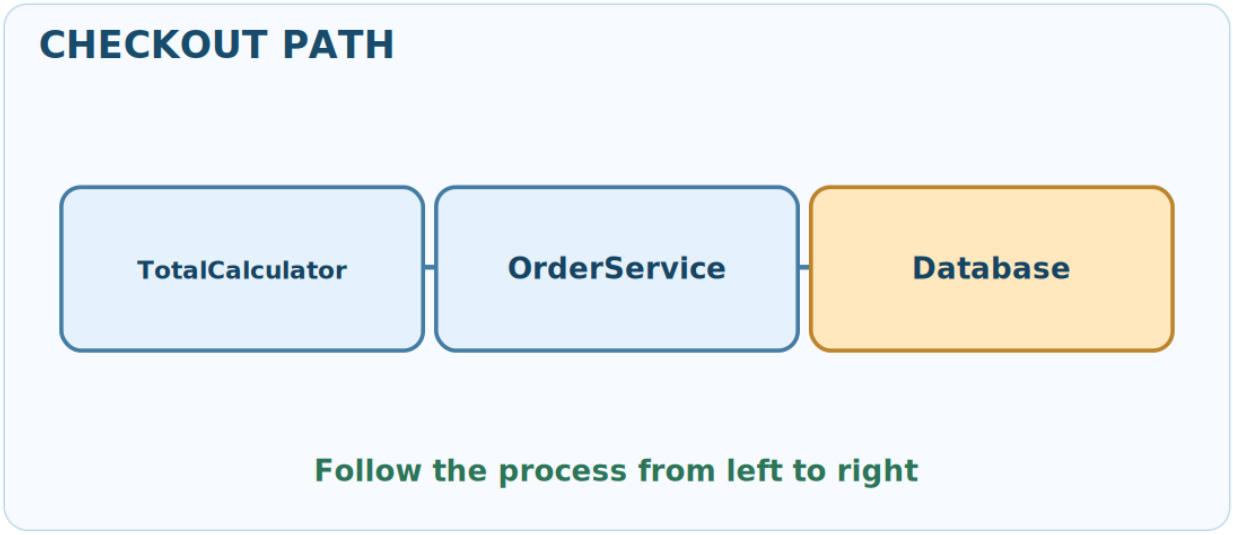
Prioritise an integration test of saving, then a system-level end-to-end checkout test.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A checkout computes the right total but fails to save orders. Which test is most relevant?

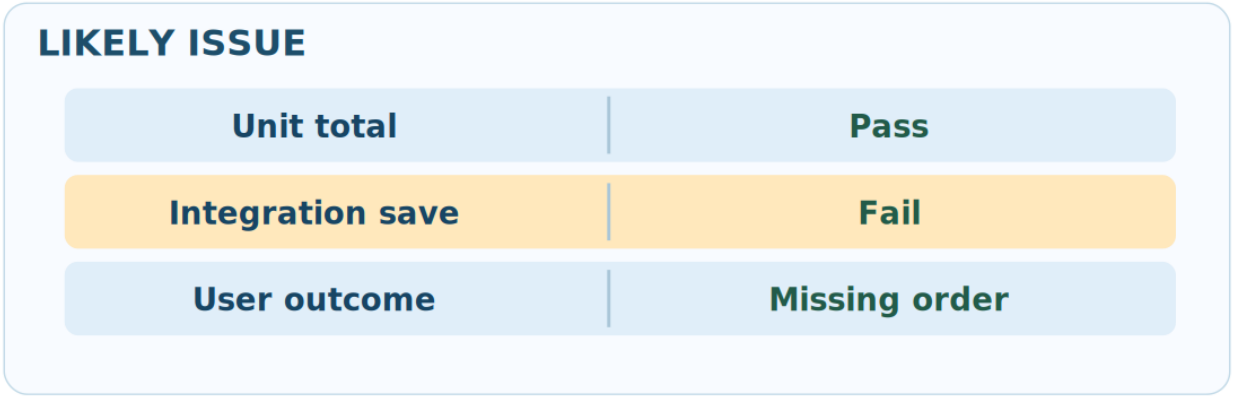
TotalCalculator already gives correct values; the failure appears when OrderService interacts with the database.



CHECKOUT PATH; TotalCalculator to OrderService to Database

A checkout computes the right total but fails to save orders. Which test is most relevant?

TotalCalculator already gives correct values; the failure appears when OrderService interacts with the database.



LIKELY ISSUE: Unit total Pass; Integration save Fail; User outcome Missing order

WE DO — GUIDED PRACTICE

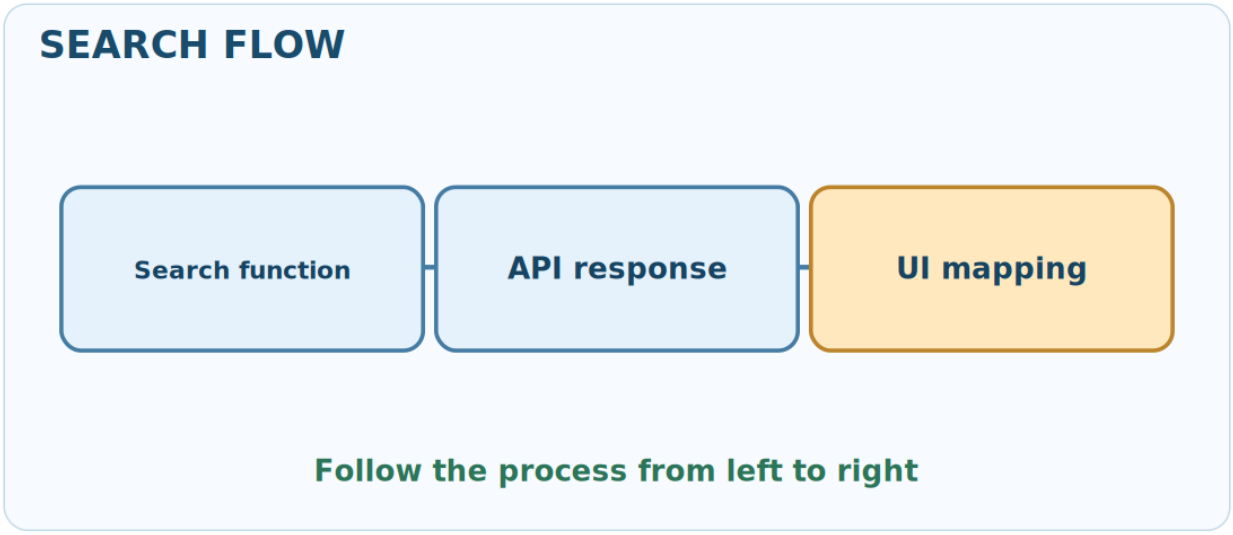
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A search function works alone but the website shows the wrong results. What should be tested?

The isolated search passes, so check the connection that maps returned IDs to displayed titles.



SEARCH FLOW; Search function to API response to UI mapping

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict first The isolated search passes, so check the connection that maps returned IDs to displayed titles.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Match each: individual ValidDate function, database interaction, full website journey, client sign-off to the appropriate test level.

Question 2

A login test passes until a new password rule is introduced, after which registration fails. Explain a sensible debugging and testing workflow.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Why is 100% execution of code lines not equivalent to proving an application correct? Suggest three further forms of evidence.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Passing unit tests alone proves that the complete application works for its users.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

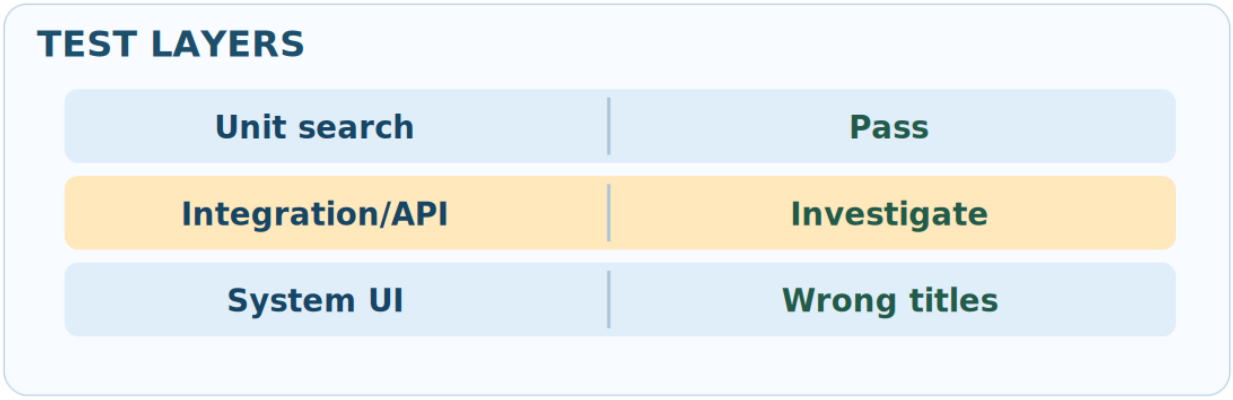
Next step: revisit the model or a relevant revision-guide section if needed.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A search function works alone but the website shows the wrong results. What should be tested?

The isolated search passes, so check the connection that maps returned IDs to displayed titles.



TEST LAYERS: Unit search Pass; Integration/API Investigate; System UI Wrong titles

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Model and explain

An integration test can expose the mapping mismatch, while a system test checks the complete user-visible result.

IMPROVE: Explain one correction to your own reasoning.